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Hosting Analytics Callback

Blindside Networks provides a hosting infrastructure for BigBlueButton, and this infrastructure can extract and summarize analytics from each BigBlueButton session.

This document describes how an LMS can receive a callback that contains a JSON file of the complete analytics when a session has finished.

This analytics from each BigBlueButton session enables the LMS provider to offer customers unified reports of student activity in both asynchronous (LMS) and synchronous (BigBlueButton) learning modes.

Overview **1**

Integrating with 3rd party applications **2**

Format of the Analytics Callback request 2

Analytics JSON Data Structure 3

Testing this callbacks **9**

Overview

Third-party applications use the BigBlueButton API to create meetings and join users.

Each time a user joins a session they generate events. These events include

- Chat events
- Talking events
- Sharing emoji
- Raising hand

By aggregating these events, the BigBlueButton server create a high-level summary of each users attendance and

- Number of times chatted
- Number of times talked
- Number of times shared emoji
- Number of times raised hand

The BigBlueButton server can also generate specific information, such as each user's

- Each response to polls
- Full Name (as passed in from the join URL)
- External UserID (as passed in from the join URL)
- isModerator (true/false)
- Join date/time
- Leave date/time
- **Total time in session**

Integrating with 3rd party applications

When a meeting ends, the BigBlueButton server processes all the events and data received from the meeting and creates a single JSON file ("data.json") of analytics for each user in the session.

To receive this data.json, the 3rd party application can pass in a meta parameter (see http://docs.bigbluebutton.org/dev/api.html#create) on the 'create' API called **meta_analytics-callback-url.**

Blindside Networks's infrastructure will use the presence of the **meta_analytics-callback-url** parameter to trigger a callback after the meeting completes, which will upload the data.json to the provided endpoint. The request will be authenticated with a JWT.

Format of the Analytics Callback request

The callback will be an HTTP POST request to the provided URL. Since the analytics contains personal data of meeting participants, the URL SHOULD be HTTPS.

The callback request will contain the following headers:

```
Content-Type: application/json
Authorization: Bearer <JWT token>
```

The JWT token must be validated. Verify the following properties:

- The HEADER would have { “typ”: “JWT”, “alg”: “HS512” }
- The PAYLOAD must contain the expiry
- The secret is the shared secret that you use for BigBlueButton API calls
- You must check the exp (expiry) parameter (the expiry time must not be in the past).

An example in Ruby would be:

```
payload = { exp: Time.now.to_i + (5 * 60) }
token = JWT.encode(payload, secret, 'HS512', type: 'JWT')
```

The body of the request will contain the JSON data structure described below. This JSON body isn’t included in the PAYLOAD in JWT.

In order for the callback to be considered successful, you must return an HTTP status code in the `2XX` range. I suggest using `200 OK` or `202 Accepted` depending on the properties of your application.
Your application may return `410 Gone` for the case where you cannot process the data because the meeting has been deleted from your application.
Redirects (HTTP status codes in the `3XX` range) will not be followed; they are treated as an error condition.

The body of the response SHOULD be empty. In the case of an error, you may return a short message in the response body with Content-Type set to text/plain for us to log to assist with debugging.

In the case of a network failure or an application failure (HTTP status codes other than `2XX` or `410`), the callback will be retried several times.

Note that the provided callback url will be ignored if it does not resolve to a public IP (e.g. if it is in a private IP range, localhost, etc.) or if it's using a reserved port number other than 80, 443, or ≥1024.

Analytics JSON Data Structure

version - data analytics structure version
meeting_id - meeting id passed from 3rd-party during CREATE API
internal_meeting_id - the meeting id within the BigBlueButton system. This is the same as the record_id for recorded meetings.
custom - any custom data needed to be passed back to callback endpoint (Optional)
data - the analytics data


```
{
	"version": "1.0",
	"meeting_id": "9674767f4f064e72afb0944b24810cb25dda4977",
	"internal_meeting_id":
"09628f80ec4d270762c8038232f149176d2f41ee-1552664944021",
	"custom": {
		"food": "pizza",
		"fruit": "apple",
		"drink": "juice"
	},
	"data": {
		"metadata": {
			"gl_listed": "false",
			"is_breakout": "false",
			"meeting_id": "9674767f4f064e72afb0944b24810cb25dda4977",
			"meeting_name": "BigBlueButton Developer's Meeting",
			"analytics_callback_url":
"https://blindside.instructure.com/api/v1/analytics?bar=baz"
		},
		"duration": 3196,
		"start": "2019-03-15 08:49:04 +0000",
		"finish": "2019-03-15 09:42:20 +0000",
		"attendees": [{
			"ext_user_id": "45",
			"name": "Sylvia Kelley",
			"moderator": true,
			"joins": ["2019-03-15 08:49:29 +0000"],
			"leaves": ["2019-03-15 09:42:20 +0000"],
			"duration": 3171.0,
			"recent_talking_time": "2019-03-15 09:42:01 +0000",
			"engagement": {
```

```
                        "chats": 3,
                        "talks": 318,
                        "raisehand": 0,
                        "emojis": 0,
                        "poll_votes": 0,
                        "talk_time": 1746
                }
        }, {
                "ext_user_id": "3245",
                "name": "Julia Franco",
                "moderator": false,
                "joins": ["2019-03-15 08:52:27 +0000", "2019-03-15 08:54:49 +0000"],
                "leaves": ["2019-03-15 08:54:49 +0000", "2019-03-15 09:42:20 +0000"],
                "duration": 2993.0,
                "recent_talking_time": "2019-03-15 09:32:09 +0000",
                "engagement": {
                        "chats": 5,
                        "talks": 2,
                        "raisehand": 0,
                        "emojis": 0,
                        "poll_votes": 5,
                        "talk_time": 3
                }
        }, {
                "ext_user_id": "gl-x6l8a62ivqgr",
                "name": "Darla Romero",
                "moderator": false,
                "joins": ["2019-03-15 08:56:02 +0000", "2019-03-15 09:12:32
+0000", "2019-03-15 09:25:55 +0000"],
                "leaves": ["2019-03-15 09:11:39 +0000", "2019-03-15 09:25:39
+0000", "2019-03-15 09:41:49 +0000"],
                "duration": 2678.0,
                "recent_talking_time": "2019-03-15 09:33:11 +0000",
                "engagement": {
                        "chats": 4,
                        "talks": 5,
                        "raisehand": 0,
                        "emojis": 0,
                        "poll_votes": 4,
                        "talk_time": 0
                }
        }, {
                "ext_user_id": "gl-ef004boifbjl",
                "name": "Howard Greer",
                "moderator": false,
                "joins": ["2019-03-15 08:59:06 +0000", "2019-03-15 08:59:39
+0000", "2019-03-15 09:00:22 +0000"],
```

"leaves": ["2019-03-15 08:59:39 +0000", "2019-03-15 09:00:19 +0000", "2019-03-15 09:00:39 +0000"],
"duration": 90.0,
"recent_talking_time": "",
"engagement": {
"chats": 0,
"talks": 0,
"raisehand": 0,
"emojis": 0,
"poll_votes": 0,
"talk_time": 0
}
}, {
"ext_user_id": "gl-3r9o2qqg9rx1",
"name": "Melinda Munoz",
"moderator": false,
"joins": ["2019-03-15 08:59:22 +0000"],
"leaves": ["2019-03-15 09:42:20 +0000"],
"duration": 2578.0,
"recent_talking_time": "2019-03-15 09:00:03 +0000",
"engagement": {
"chats": 4,
"talks": 1,
"raisehand": 0,
"emojis": 0,
"poll_votes": 5,
"talk_time": 1
}
}, {
"ext_user_id": "3432",
"name": "**Loretta Wade**",
"moderator": false,
"joins": ["2019-03-15 09:01:14 +0000", "2019-03-15 09:29:52 +0000"],
"leaves": ["2019-03-15 09:29:49 +0000", "2019-03-15 09:42:20 +0000"],
"duration": 2463.0,
"recent_talking_time": "",
"engagement": {
"chats": 2,
"talks": 0,
"raisehand": 0,
"emojis": 0,
"poll_votes": 3,
"talk_time": 0
}
}, {

"ext_user_id": "gl-urs3p3bsluvm",
"name": "Wilfred Neal",
"moderator": false,
"joins": ["2019-03-15 09:01:19 +0000", "2019-03-15 09:11:21 +0000", "2019-03-15 09:25:39 +0000", "2019-03-15 09:33:41 +0000", "2019-03-15 09:41:49 +0000"],
"leaves": ["2019-03-15 09:09:29 +0000", "2019-03-15 09:23:19 +0000", "2019-03-15 09:33:39 +0000", "2019-03-15 09:41:49 +0000", "2019-03-15 09:42:20 +0000"],
"duration": 2207.0,
"recent_talking_time": "",
"engagement": {
"chats": 1,
"talks": 0,
"raisehand": 0,
"emojis": 0,
"poll_votes": 2,
"talk_time": 0
}
}, {
"ext_user_id": "gl-u072ra49wamf",
"name": "Hubert Buchanan",
"moderator": false,
"joins": ["2019-03-15 09:02:05 +0000"],
"leaves": ["2019-03-15 09:42:09 +0000"],
"duration": 2404.0,
"recent_talking_time": "2019-03-15 09:34:30 +0000",
"engagement": {
"chats": 3,
"talks": 16,
"raisehand": 0,
"emojis": 0,
"poll_votes": 5,
"talk_time": 43
}
}, {
"ext_user_id": "gl-w2ju856ikegp",
"name": "Constance Rodriguez",
"moderator": false,
"joins": ["2019-03-15 09:05:01 +0000", "2019-03-15 09:08:09 +0000"],
"leaves": ["2019-03-15 09:07:29 +0000", "2019-03-15 09:42:19 +0000"],
"duration": 2198.0,
"recent_talking_time": "2019-03-15 09:42:17 +0000",
"engagement": {
"chats": 6,

```
                        "talks": 15,
                        "raisehand": 0,
                        "emojis": 0,
                        "poll_votes": 4,
                        "talk_time": 23
                }
        }, {
                "ext_user_id": "gl-rkwcqpxjpykl",
                "name": "Leroy Taylor",
                "moderator": false,
                "joins": ["2019-03-15 09:06:43 +0000", "2019-03-15 09:08:32
+0000"],
                "leaves": ["2019-03-15 09:08:26 +0000", "2019-03-15 09:42:09
+0000"],
                "duration": 2120.0,
                "recent_talking_time": "",
                "engagement": {
                        "chats": 5,
                        "talks": 0,
                        "raisehand": 0,
                        "emojis": 0,
                        "poll_votes": 3,
                        "talk_time": 0
                }
        }, {
                "ext_user_id": "gl-mvjbefisdkcl",
                "name": "Everett Baker",
                "moderator": false,
                "joins": ["2019-03-15 09:19:24 +0000"],
                "leaves": ["2019-03-15 09:42:19 +0000"],
                "duration": 1375.0,
                "recent_talking_time": "2019-03-15 09:41:48 +0000",
                "engagement": {
                        "chats": 5,
                        "talks": 26,
                        "raisehand": 0,
                        "emojis": 0,
                        "poll_votes": 2,
                        "talk_time": 98
                }
        }],
        "files": ["default.pdf"],
        "polls": [{
                "id":
"d2d9a672040fbde2a47a10bf6c37b6a4b5ae187f-1552664944032/2/1552665888981",
                "published": true,
                "options": ["Yes", "No"],
```

```
                "votes": {
                        "34": "Yes",
                        "44": "Yes",
                        "gl-3r9o2qqg9rx1": "Yes",
                        "gl-x6l8a62ivqgr": "Yes",
                        "gl-urs3p3bsluvm": "Yes",
                        "gl-fis0njvz531y": "Yes"
                },
                "start": "2019-03-15 09:04:48 +0000"
        }, {
                "id":
"d2d9a672040fbde2a47a10bf6c37b6a4b5ae187f-1552664944032/2/1552666421691",
                "published": true,
                "options": ["Yes", "No"],
                "votes": {
                        "gl-w2ju856ikegp": "Yes",
                        "gl-u072ra49wamf": "Yes",
                        "gl-yxo1vubwl6kk": "Yes",
                        "gl-fis0njvz531y": "Yes",
                        "gl-x6l8a62ivqgr": "Yes",
                        "gl-3r9o2qqg9rx1": "Yes"
                },
                "start": "2019-03-15 09:13:41 +0000"
        }, {
                "id":
"d2d9a672040fbde2a47a10bf6c37b6a4b5ae187f-1552664944032/3/1552667336280",
                "published": true,
                "options": ["Yes", "No"],
                "votes": {
                        "gl_rkwcqpxjpykl": "Yes",
                        "gl-u072ra49wamf": "Yes",
                        "gl-x6l8a62ivqgr": "Yes",
                        "gl-3r9o2qqg9rx1": "Yes",
                        "gl_mvjbefisdkcl": "Yes",
                        "gl-yxo1vubwl6kk": "Yes",
                        "gl-w2ju856ikegp": "Yes"
                },
                "start": "2019-03-15 09:28:56 +0000"
        }, {
                "id":
"d2d9a672040fbde2a47a10bf6c37b6a4b5ae187f-1552664944032/2/1552667462355",
                "published": true,
                "options": ["Yes", "No"],
                "votes": {
                        "gl-w2ju856ikegp": "Yes",
                        "gl-fis0njvz531y": "Yes",
                        "gl-urs3p3bsluvm": "Yes",
```

```
                        "gl_mvjbefisdkcl": "Yes",
                        "gl-u072ra49wamf": "Yes",
                        "gl_rkwcqpxjpykl": "Yes",
                        "gl-yxo1vubwl6kk": "Yes",
                        "gl-3r9o2qqg9rx1": "Yes"
                    },
                    "start": "2019-03-15 09:31:02 +0000"
            }, {
                    "id":
"d2d9a672040fbde2a47a10bf6c37b6a4b5ae187f-1552664944032/2/1552667980092",
                    "published": true,
                    "options": ["Yes", "No"],
                    "votes": {
                        "gl-x6l8a62ivqgr": "No",
                        "gl-3r9o2qqg9rx1": "No",
                        "gl-w2ju856ikegp": "No",
                        "gl-u072ra49wamf": "No",
                        "gl-yxo1vubwl6kk": "No",
                        "gl_rkwcqpxjpykl": "No"
                    },
                    "start": "2019-03-15 09:39:40 +0000"
            }]
        }
}
```


Testing this callbacks

Next Steps:

1. Setup an endpoint where you can receive an Analytics Callback from Blindside Networks. Provide us a URL to that endpoint.
2. Blindside Networks will configure a test account using that endpoint and provide you a shared secret to validate the JWT token.
3. Customers can start testing using this test account. When a session finishes on that test account, you'll receive callbacks at the URL with a signed JWT Token (using the shared secret) along with the json.data meeting analytics. a

For help

If you need help on the implementation, please email us at support@blindsidenetworks.com